

Towards Trustworthy Automated Essay Scoring: Explainable Transformers and Efficient Fine-Tuning Strategies

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Abstract

Artificial Intelligence allowed instructors to automate the grading of students' work. Many studies show the high performance of neural networks in automated grading. However, their "Blackbox" nature reduces the educators' trust in adopting it in the educational context. In this study, we propose a framework for automated essay scoring using Transformer-based models and Explainable AI (XAI). We evaluated the five models on two different benchmarking datasets, ASAP 2.0 and Learning Agency Lab, on models: Linear Regression, Random Forest, BiLSTM with Attention, BERT, and FLAN-T5. Our studies show the BERT model achieved a Quadratic Weighted Kappa (QWK) score of 0.8431, which is 19.5% better than traditional machine learning models. Moreover, the study also shows the cross-prompt generalization with 70.3% QWK degradation in leave-one-prompt-out evaluation, but it improves when training data contains diverse prompts. For resource optimization, we apply Parameter-Efficient Fine-Tuning (PEFT) using LoRA on FLAN-T5. This approach achieves 86.2% of BERT's Pearson correlation while training only 1.8% of parameters. Additionally, we integrated LIME and SHAP to provide pedagogically transparent explanations across all model families. Overall, the results show that training strategies play a more important role than model architecture, and parameter-efficient methods provide practical solutions for real educational systems.

Keywords: Automated Essay Scoring, Explainable AI, Transformers, BERT, FLAN-T5, LoRA, SHAP, LIME, Cross-Prompt Generalization

I. INTRODUCTION

The Automated Essay Scoring (AES) has emerged as a more important area in the research community for giving more consistent, reliable, accurate, and timely assessment of students' work. The traditional manual grading has flaws like being time-consuming, subjective interpretation, and evaluator bias. As the amount of work has been increasing in the digital learning world, it is often difficult to check each student's work manually. To solve these issues, many researchers have tried to automate this process by leveraging Artificial Intelligence (AI). It provides a practical solution for supporting instructors while maintaining evaluation efficiency [1, 2]. Beyond automated grading, machine learning approaches have also been deployed in broader educational contexts, such as predicting overall student academic performance [3]. In the very beginning, Machine Learning (ML) models like linear regression, support vector machines, and random forests were used to automate this grading. These models used features such as word frequency, syntactic complexity, and lexical diversity to estimate essay quality, which shows good performance but hardly captures the deeper semantic relationships and discourse structures within essays [4, 5].

To enhance deeper semantic relations and structure, modern Deep Learning (DL) architectures like BERT have shown strong performance in capturing contextual and semantic relationships in text through bidirectional language representations [6]. These DL models have shown high performance comparable to human raters on benchmarked datasets. However, regardless of their high performance, DL models are used by many educators because of their blackbox nature, which reduces the transparency and trust of evaluators to adopt it real educational settings [8–12].

A. Contributions

The key contributions of this work include:

1. A unified progressive experimental framework comparing five model families (LR, RF, BiL-STM, BERT/RoBERTa, FLAN-T5) on identical datasets under consistent evaluation protocols.
2. Empirical demonstration that training methodology (warm-up, weight decay, mixed-precision) accounts for a 34.3% QWK improvement using the same BERT architecture.
3. First systematic investigation of LoRA and QLoRA for AES, achieving competitive performance with only 1.8% trainable parameters.
4. A Cross-prompt generalization analysis, which reveals 70.3% QWK degradation and its mitigation through diverse-prompt training.
5. Integrated multi-model SHAP and LIME explainability analysis providing pedagogical transparency.

II. RELATED WORK

Recent studies have shown that automated essay scoring has been performed in various approaches using machine learning, neural networks, and transformer-based architectures. Kansagra et al. [21] conducted a comprehensive survey on machine learning and neural networks for automating essay scoring and showed, in their results, that the hybrid approach combining both architectures often yields the best trade-off between performance and interpretability.

Oye et al. [13] performed a comprehensive comparative analysis and found that the shape technique of explainable Artificial Intelligence (XAI) gives more consistent explanations across different types of essays. On the other hand, Bontempelli et al. [16] introduced the neural additive models (NAMs) technique for explaining short answers, eliminating the need for post-hoc methods. Poulton et al. [7] show that the attention visual technique of XAI aligns with the human grading evidence, offering a natural way to transparency. Manna et al. [14] have proof from user studies that XAI-enhanced grading systems increase student trust significantly and perceived fairness. Recently, Qiu et al. [17] used retrieval-augmented generation (RAG) with large language models (LLMs) for generating the rubric-based scores and explanations. Regardless of these advances, the current existing studies often measure and evaluate the individual model families in isolation, not with a hybrid, and often use high resources, lacking the investigation parameter-efficient fine-tuning for AES, and hardly apply the XAI on various models under the unified controlled conditions. Our study bridges this gap by providing a framework that uses traditional ML, LSTMs, BERT, and FLAN-T5, all evaluated with integrated SHAP/LIME explainability. Figure 1 shows the overall research methodology, showing the three-phase progressive experimental design

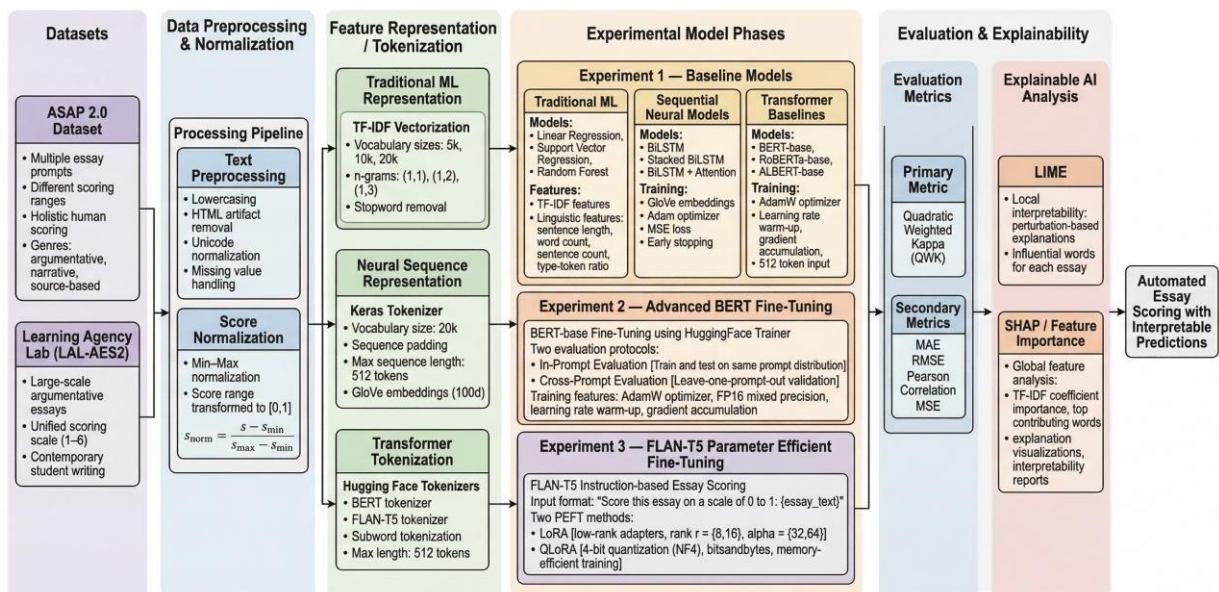


Figure 1: Overall research methodology showing the three-phase progressive experimental design

III. METHODOLOGY

This section explains the methodology that we followed in this study. The study followed typically three phases, each phase is built upon the findings from the previous one. Firstly, in phase one, we evaluated the traditional ML models on the automated essay score to have the baseline results to compare the rest. We used Linear regression and Random forest, along with recurrent neural networks (BiLSTM with attention), and standard transformer models (BERT, RoBERTa, ALBERT) under both in-prompt and cross-prompt settings. These models are used as per the literature findings. Moreover, in the second Phase, we focused on advanced fine-tuning strategies to train the BERT model entirely while incorporating techniques such as learning rate warm-up, weight decay, mixed-precision training, and gradient accumulation. Further, in the third phase, we leverage the parameter-efficient fine-tuning (PEFT) for the FLAN-T5 model using LoRA and QLoRA to achieve competitive scoring performance while drastically reducing the number of trainable parameters.

A. Research Design

The study adopts a three-phase progressive design as illustrated in Fig. 1:

1. **Experiment 1 (Baselines):** Traditional ML (LR, Support Vector Regression (SVR), RF), BiL-STM variants, and Transformer baselines (BERT, RoBERTa [23], ALBERT) with cross-prompt generalization analysis.
2. **Experiment 2 (Advanced Fine-Tuning):** BERT with learning rate warm-up, weight decay, 16-bit floating-point mixed precision (FP16), and gradient accumulation under in-prompt and cross-prompt protocols.
3. **Experiment 3 (PEFT):** FLAN-T5 with LoRA and QLoRA across multiple rank/alpha configurations.

B. Datasets

Two benchmark datasets are used: **ASAP 2.0** (multi-prompt, multi-genre student essays with prompt-specific scoring rubrics) [18] and **Learning Agency Lab (LAL)** (large-scale argumentative essays with a unified 1–6 scoring range) [19]. All scores are normalized to $[0, 1]$ using

min-max normalization:

$$s_{\text{norm}} = \frac{s - s_{\min}}{s_{\max} - s_{\min}} \quad (1)$$

C. Model Architectures

Linear Regression learns a direct mapping from TF-IDF features:

$$\hat{y} = \mathbf{w}^T \mathbf{x} + b \quad (2)$$

Random Forest operates on combined TF-IDF and linguistic features (average sentence length, Type-Token Ratio, word count, sentence count).

BiLSTM with Attention computes a learned weighted aggregation over hidden states:

$$e_t = \tanh(\mathbf{W}_a \mathbf{h}_t + \mathbf{b}_a), \quad \alpha_t = \frac{\exp(e_t)}{\sum_j \exp(e_j)}, \quad \mathbf{c} = \sum_t \alpha_t \mathbf{h}_t \quad (3)$$

BERT-base-uncased [22] (110M parameters) uses the [CLS] token through a linear regression head. **FLAN-T5-base** [25], which builds upon the foundational T5 framework [26], frames scoring as text-to-text generation: "Score this essay on a scale of 0 to 1: {essay}".

LoRA [27] injects trainable low-rank matrices into frozen attention layers:

$$\mathbf{W}' = \mathbf{W}_0 + \Delta \mathbf{W} = \mathbf{W}_0 + \mathbf{B} \mathbf{A} \quad (4)$$

where $\mathbf{B} \in \mathbb{R}^{d \times r}$, $\mathbf{A} \in \mathbb{R}^{r \times k}$ are trainable adapters. **QLoRA** [28] extends LoRA with 4-bit NF4 quantization. Table 1 shows the PEFT configurations.

Table 1: FLAN-T5 PEFT Experiment Configurations

Exp.	Method	Rank (r)	Alpha (α)	Dropout	Batch
3.1	LoRA	8	32	0.05	8
3.2	QLoRA	8	32	0.05	4
3.3	LoRA	16	64	0.05	8
3.4	QLoRA	16	64	0.05	4

D. Evaluation Metrics

Performance is assessed using Quadratic Weighted Kappa (QWK), Pearson Correlation, MAE and RMSE:

$$\kappa = 1 - \frac{\sum_{i,j} w_{ij} O_i}{\sum_{i,j} w_{ij} E_i}, \quad w_{ij} = \frac{(i-j)^2}{(N-1)^2} \quad (5)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad r = \frac{\sum_{i=1}^n (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2 \sum_{i=1}^n (\hat{y}_i - \bar{\hat{y}})^2}} \quad (6)$$

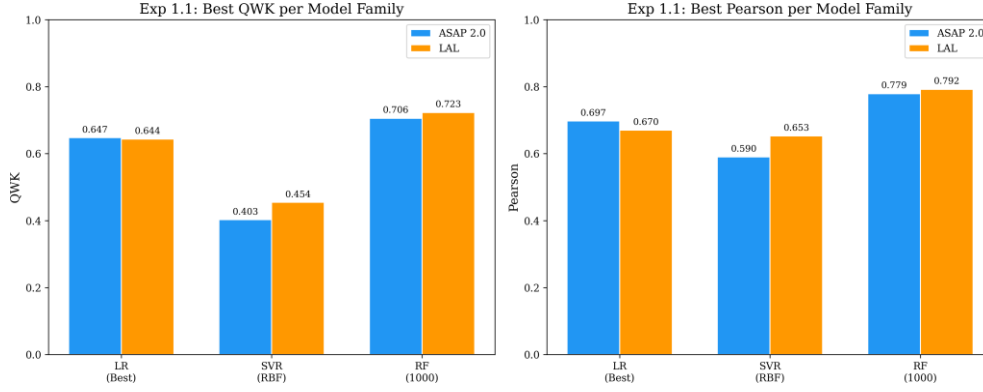
E. Explainability Framework

Post-hoc XAI analysis uses **LIME** [29] for local instance-level explanations (perturbing input words, fitting surrogate linear models) and **SHAP** [20] for global feature importance through coefficient analysis.

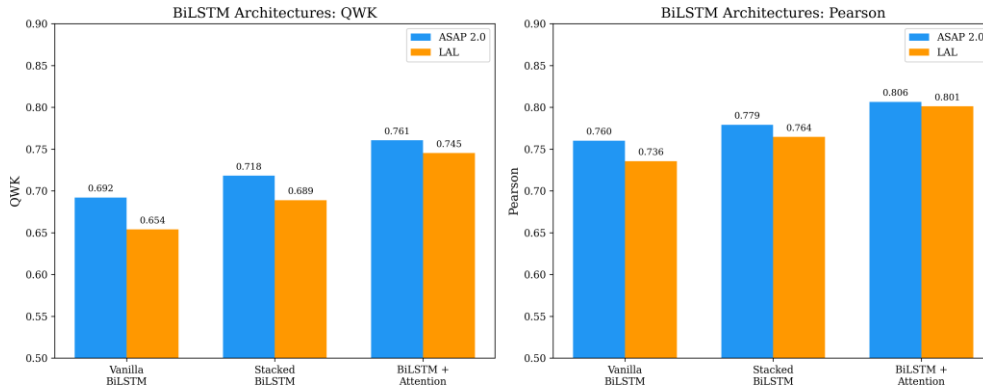
IV. EXPERIMENTAL RESULTS

A. Experiment 1: Traditional ML and LSTM Baselines

Table 2 presents the best-performing traditional ML and LSTM models. Random Forest with combined TF-IDF and linguistic features established the strongest traditional ML baseline



(a) Traditional ML comparison (ASAP)



(b) BiLSTM architectural comparison

Figure 2: Performance comparison of baseline models

(QWK = 0.7055 on ASAP). BiLSTM with Attention surpassed all ML models (QWK = 0.7606, +7.8% over RF), as shown in Figure 2, confirming that selective attention captures compositional essay semantics beyond bag-of-words representations.

Table 2: Best-Performing Baseline Models Across Datasets

Model	Dataset	QWK	Pearson	MAE	RMSE
LR (v=20K, n=1,2)	ASAP	0.6474	0.6971	0.5800	0.7307
LR (v=5K, n=1,2)	LAL	0.6437	0.6698	0.6509	0.8180
RF (1000 Trees)	ASAP	0.7055	0.7790	0.4820	0.6270
RF (1000 Trees)	LAL	0.7229	0.7921	0.4872	0.6385
Vanilla BiLSTM	ASAP	0.6920	0.7601	0.5206	0.6760
Stacked BiLSTM	ASAP	0.7181	0.7789	0.5032	0.6510
BiLSTM + Attention	ASAP	0.7606	0.8063	0.4775	0.6141
BiLSTM + Attention	LAL	0.7452	0.8012	0.4874	0.6260

Key Takeaway: Random Forest with combined features established the strongest ML baseline (QWK = 0.7055), while BiLSTM with Attention surpassed all ML models by 7.8%, confirming that learned attention over sequential hidden states captures compositional essay semantics beyond bag-of-words representations.

B. Cross-Prompt Generalization Study

Table 3 shows severe generalization deficits in the leave-one-prompt-out evaluation. The average cross-prompt QWK dropped to 0.1856—a 70.3% degradation—with heterogeneous

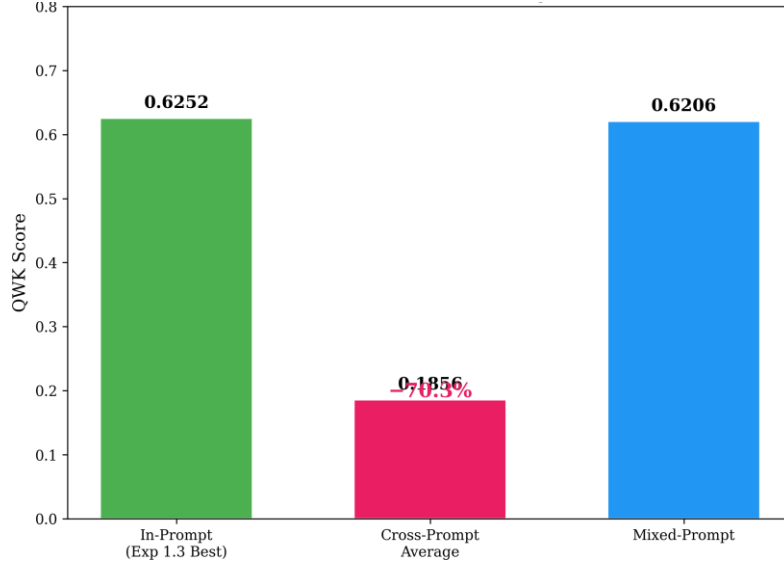


Figure 3: Performance degradation from in-prompt to cross-prompt evaluation

performance across prompts (QWK range: 0.0186–0.3389). Mixed-prompt training recovered performance to QWK = 0.6206, demonstrating that diverse-prompt exposure is both necessary and sufficient for learning generalizable scoring features. Figure 3 shows Performance degradation from in-prompt to cross-prompt evaluation.

Table 3: Cross-Prompt Generalization Analysis

Evaluation Protocol	QWK	Pearson	Δ QWK	Degrad. (%)
In-Prompt (Best)	0.6252	0.6726	—	—
Cross-Prompt Average	0.1856	0.3782	-0.4396	-70.3%
Mixed-Prompt (All)	0.6206	0.7147	-0.0046	-0.7%

Key Takeaway: Cross-prompt evaluation reveals a critical 70.3% QWK degradation, but mixed-prompt training recovers performance to within 0.7% of in-prompt results, confirming that diverse-prompt exposure is both necessary and sufficient for generalizable scoring.

C. Experiment 2: Advanced BERT Fine-Tuning

Table 4 shows BERT fine-tuning results with advanced training strategies. The cross-prompt model (QWK = 0.8431) slightly outperformed the in-prompt model, attributable to the regularization effect of diverse-prompt training. Moreover, Table 5 presents per-fold cross-prompt results. ASAP prompts achieved highly consistent QWK (0.8348–0.8489, $\sigma = 0.006$), while the LAL prompt scored lower (QWK = 0.7675) due to cross-dataset distributional mismatch.

Table 4: BERT Fine-Tuning Results: In-Prompt vs. Cross-Prompt (Epoch 3)

Protocol	QWK		Person	Δ vs. Baseline
In-Prompt	0.8397	0.3858	0.8761	+19.0%
Cross-Prompt	0.8431	0.4115	0.8871	+19.5%

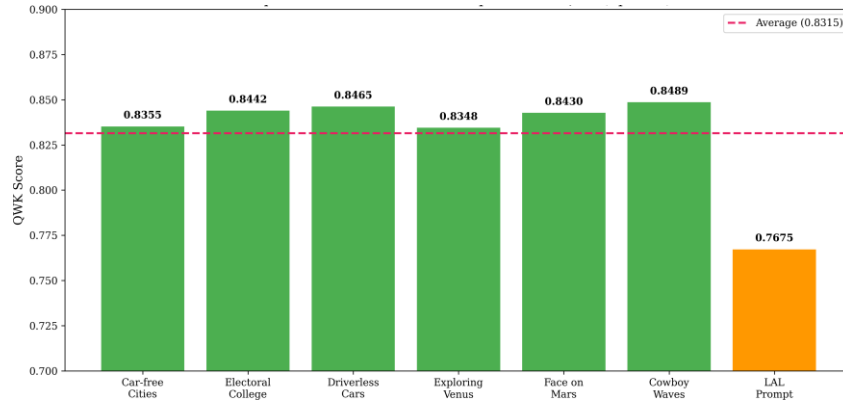


Figure 4: Per-fold QWK for BERT cross-prompt evaluation (Epoch 3)

Table 5: BERT Cross-Prompt Per-Fold Results at Epoch 3

Held-Out Prompt	QWK	MAE	Pearson
Car-free Cities	0.8355	0.4168	0.8739
Does the Electoral College Work?	0.8442	0.4014	0.8794
Driverless Cars	0.8465	0.4168	0.8845
Exploring Venus	0.8348	0.4164	0.8760
The Face on Mars	0.8430	0.4200	0.8853
“A Cowboy Who Rode the Waves”	0.8489	0.4072	0.8880
LAL Prompt	0.7675	0.4944	0.8277
Average	0.8315	0.4247	0.8735

Key Takeaway: Advanced fine-tuning strategies (warm-up, weight decay, FP16, gradient accumulation) yield a 34.3% QWK improvement over the same BERT architecture with basic training, demonstrating that training infrastructure is as impactful as model selection. Figure 4 presents the cross-prompt evaluation.

D. Experiment 3: FLAN-T5 with PEFT

Table 6 presents the PEFT fine-tuning results. LoRA ($r=8$) achieved the best Pearson (0.7645), outperforming the higher-rank LoRA ($r=16$), confirming implicit regularization from low-rank constraints. QLoRA achieved 1.8 $\times$ higher throughput while maintaining competitive accuracy.

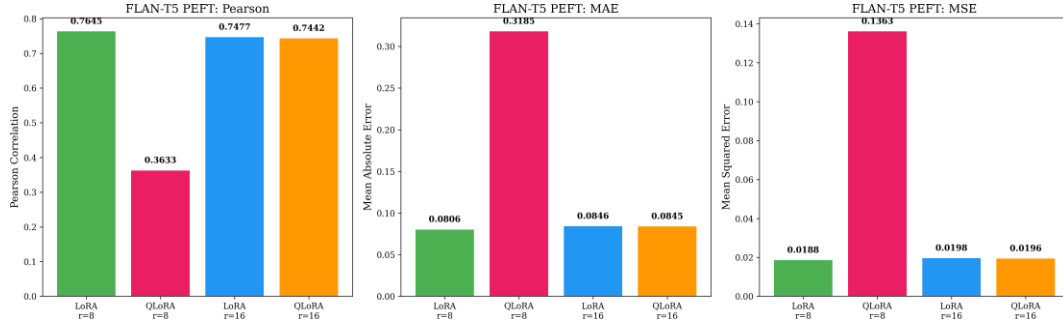
Table 6: FLAN-T5 PEFT Results and Computational Efficiency

Exp.	Method	Pearson	MSE	MAE	Eval Time (s)	Samples/s
3.1	LoRA ($r=8$)	0.7645	0.0188	0.0806	836.14	2.958
3.2	QLoRA ($r=8$)	0.3633	0.1363	0.3185	468.00	5.287
3.3	LoRA ($r=16$)	0.7477	0.0198	0.0846	736.18	3.359
3.4	QLoRA ($r=16$)	0.7442	0.0196	0.0845	459.83	5.378

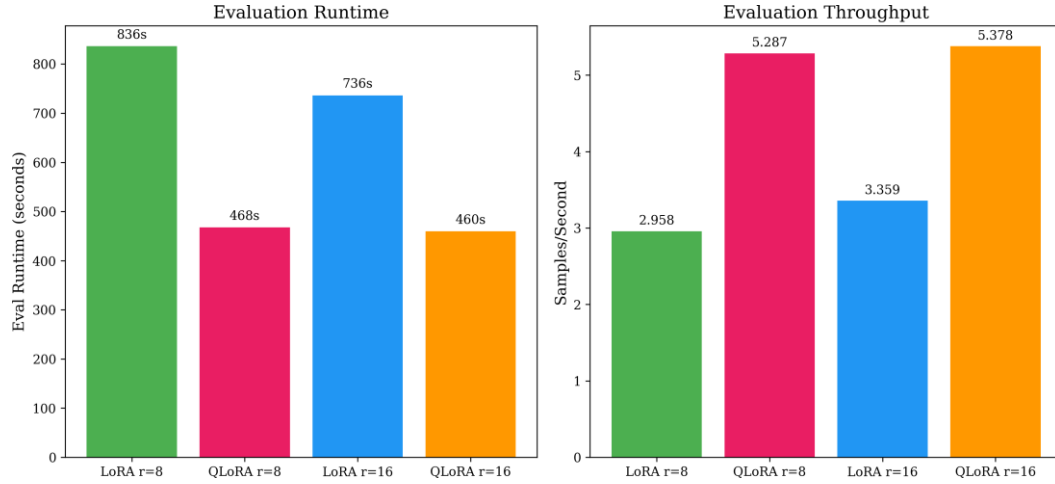
Key Takeaway: LoRA ($r=8$) achieves the best Pearson (0.7645) among all PEFT configurations, outperforming higher-rank LoRA ($r=16$) and confirming implicit regularization benefits. QLoRA ($r=8$) underperforms significantly (Pearson = 0.3633), suggesting that aggressive quantization at low rank destabilizes learning. QLoRA ($r=16$) offers the best efficiency trade-off with 1.8 $\times$ higher throughput while maintaining 97.3% of LoRA’s Pearson.

E. Cross-Experiment Comparison

Table 7 consolidates results across all experimental phases.



(a) LoRA vs. QLoRA performance



(b) Evaluation throughput comparison

Figure 5: FLAN-T5 PEFT performance and efficiency analysis**Table 7:** Grand Comparison: Best Model from Each Phase

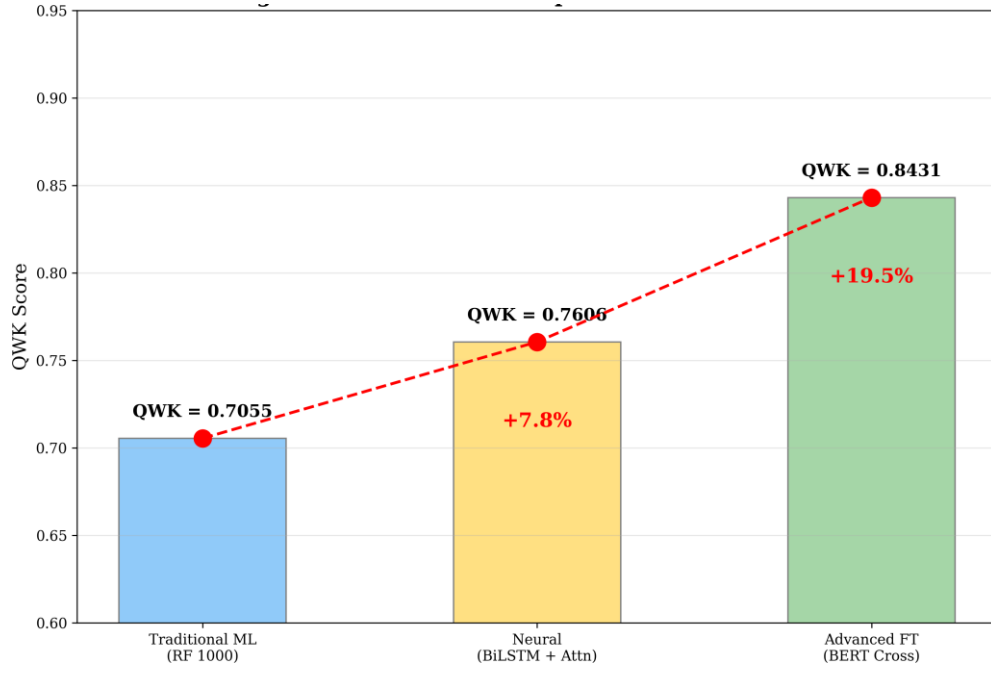
Phase	Best Model	QWK	Pearson	Trainable	Improv.
1.1 (ML)	RF (1000 Trees)	0.7055	0.7790	~50M+	—
1.2 (LSTM)	BiLSTM + Attention	0.7606	0.8063	~1M	+7.8%
1.3 (Trans.)	RoBERTa (lr=3e-5)	0.6252	0.6726	~110M	—
2 (BERT Adv.)	BERT Cross-Prompt	0.8431	0.8871	~110M	+19.5%
3 (PEFT)	LoRA (r=8)	— [†]	0.7645	~2M	—

[†]QWK not computed for FLAN-T5 (Seq2Seq generation paradigm). And Table 8 shows Parameter efficiency analysis, while Figure 5 evaluates FLAN-T5 PEFT performance and efficiency analysis

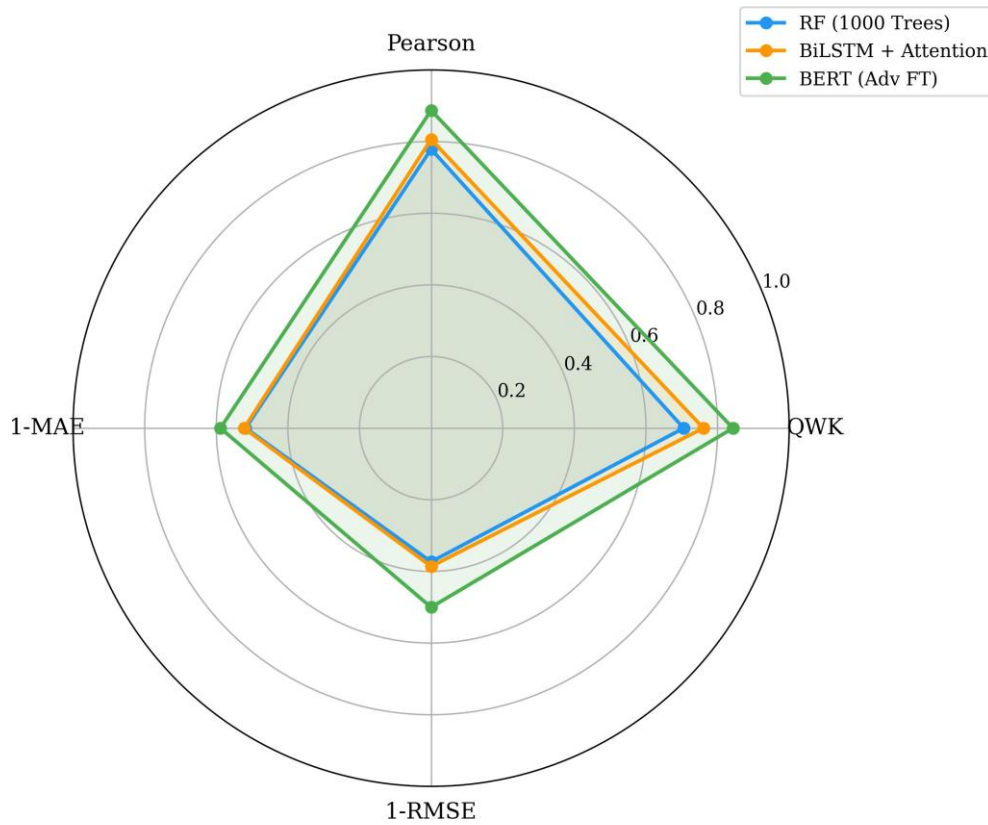
Table 8: Parameter Efficiency Analysis

Model	Pearson	Trainable Params	Pearson/M Params
BERT (Full FT)	0.8871	~110M	0.0081
BiLSTM + Attention	0.8063	~1M	0.8063
FLAN-T5 + LoRA (r=8)	0.7645	~2M	0.3823
FLAN-T5 + QLoRA	0.7442	~2M	0.3721

FLAN-T5 + LoRA achieves 86.2% of BERT's Pearson (0.7645/0.8871) while training only 1.8% of parameters (2M/110M). Figure 6 presents the Grand synthesis: from baselines to advanced Transformer models



(a) Progressive QWK improvement



(b) Multi-metric performance profile

Figure 6: Grand synthesis: from baselines to advanced Transformer models

F. Explainability Analysis

SHAP and LIME analyses were applied across all model families. Fig. 7 presents representative XAI outputs.

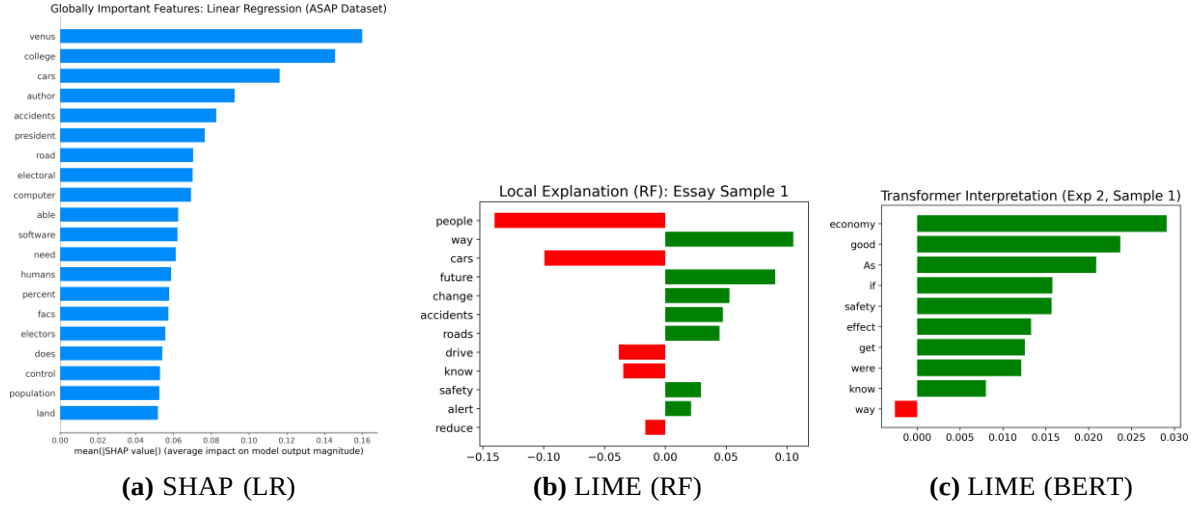


Figure 7: XAI analysis across model families: (a) SHAP global importance for Linear Regression, (b) LIME local explanation for Random Forest, (c) LIME explanation for fine-tuned BERT

As shown in Figure 7, the analysis reveals a progressive shift in model reasoning: **Random Forest** heavily relies on surface-level content keywords, and **BERT** attends to semantic coherence and argument structure through contextual phrase spans. Moreover, **FLAN-T5** exhibits sensitivity to instruction prefixes and structural positional cues.

V. DISCUSSION

The experimental results showed several important insights across the three phases of this study. The unique BERT architecture shows very different results. It heavily depends on the training settings, like in the initial phase of fine-tuning, which gave limited baseline results, then advanced fine-tuning shows a huge improvement of 34.3% improvement in QWK=0.8413% with warm-up scheduling, weight decay, FP16 mixed precision, and gradient accumulation settings. This demonstrates that the training infrastructure is as important as the model selection, a finding that has important practical implications for AES deployment. Cross-prompt generalization analysis exposes structural fairness concerns. The observation of 70.3% QWK degradation in the leave-one-prompt-out evaluation highlights the risk of immediate specific overfitting in AES systems. However, we resolved this issue with mix prompt training effectively. BERT with cross-prompt achieved QWK=0.8431 than the in-prompt model (QWK = 0.8397), with highly consistent per-fold performance ($\sigma = 0.006$). It is shown that diverse prompt exposure allows the model to learn general scoring features rather than specific patterns. With PEFT(parameter-efficient fine-tuning), FLAN-T5 offers great performance with limited resource consumption. LoRA ($r=8$) outperforms LoRA ($r=16$) by confirming the implicit regularization with low-level constraints, while QLoRA ($r=16$) achieves 97.3% of the Pearson correlation of LoRA, enabling deployment on consumer-grade 8 GB GPUs. However, at low rank ($r=8$) QLoRA exhibited degraded performance (Pearson = 0.3633), indicating that the aggressive amount of learning combined with low rank adaptation can destabilize the learning process. LIME analysis further revealed that changes in model complexity *what* is evaluated: Random Forest evaluates surface-level keywords, BERT evaluates structural keywords, and FLAN-T5 evaluates instruction-guided structural features which showing that the architecture itself determines the structure.

VI. ETHICAL CONSIDERATIONS

In the context of Automated Essay Scoring (AES), the main concern is bias and fairness, particularly in cross-prompt generalization (e.g., leave-one-prompt-out evaluation). Our study

shows that models trained on specific prompts tend to exhibit significant performance degradation (approximately 70% QWK drop) when tested on unseen prompts. This negatively affects student assessment, especially on underrepresented topics.

We propose mixed-prompt training as a solution to mitigate this bias; however, careful attention must be paid to ensuring diverse and representative training data. Furthermore, the integration of explainable AI (XAI) techniques such as SHAP and LIME serves as both a technical and ethical safeguard. These methods allow teachers and examiners to audit model decisions for potential biases prior to deployment.

It is important that AES systems be used as decision-support tools rather than as sole evaluators, and that regular fairness audits be conducted across demographic groups and prompt categories [30, 33].

VII. CONCLUSION

In this study, we have proposed a unified Transformer-driven AES framework with integration of XAI techniques for model transparency across five model families evaluated on two benchmark datasets. With advanced fine-tuning BERT to overcome biasness, we have achieved QWK = 0.8431, a 19.5% improvement over ML baselines, with training methodology alone accounting for a 34.3% QWK improvement. This demonstrates that training infrastructure is as critical as model selection. In cross-prompt testing model performs degrade by 70.3% of QWK. We mitigated this issue through diverse-prompt training. Moreover, FLAN-T5 with LoRA ($r=8$) achieved 86.2% of BERT's Pearson correlation while training only 1.8% of parameters, offering a resource-efficient deployment pathway. LIME and SHAP analyses showed the presence of progressive movements in the reasoning from keywords to compositional semantics within different model families, offering pedagogical transparency. This study will be expanded in the future, to include multilingual AES extension, larger generative models for joint scoring and feedback generation, trait-based analytic scoring, and full PEFT ablation studies on additional models of AES.

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